THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY, PATIALA

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

END SEMESTER TEST

MCA- 1st Year	**Course Code:** MCA103
MMs: 40	**Course:** Operating System
Date: 10 Dec 2024	**Faculty:** Dr. Shubhra Dwivedi

Instructions: Attempt any five questions and all sub parts like (a) & (b) for each question at one place. Do mention Page No. of your attempt at front page of the answer sheet. Assume missing data (if any).			**CO**	**Level**
Q.1. (a)	Briefly explain, what are conditions for deadlock to occur? In a system, following state of processes and resources are given: R1→ P1, P1→ R2, P2→ R3, R2→ P2, R3→ P3, P3→ R4, P4→ R3, R4→ P4, P4→ R1, R1→ P5. **Solve the following queries.** I. Draw suitable Resource Allocation Graph II. Draw corresponding wait-for-graph III. Identify is their deadlock in system or not.	4	CO3	L3
(b)	Explain what a Translation Lookaside Buffer (TLB) is and its role in a paging system with a suitable diagram. A paging scheme uses a TLB. The effective memory access takes 160 ns and a main memory access takes 100 ns. What is the TLB access time (in ns) if the TLB hit ratio is 60% and there is no page fault?	4	CO2	L4
Q.2. (a)	Consider the following snapshot of a system. *Allocation* / *Max* (A B C D): T_0: 3014 / 5117 T_1: 2210 / 3211 T_2: 3121 / 3321 T_3: 0510 / 4612 T_4: 4212 / 6325 Using the banker's algorithm, determine whether or not each of the given states is unsafe. If the state is safe, illustrate the order in which the threads (T) may complete. Otherwise, illustrate why the state is unsafe. I. Available = (0, 3, 0, 1) II. Available = (1, 0, 0, 2)	4	CO5	L3
(b)	Consider a disk with 16 platters, 2 surfaces per platter, 1k tracks per surface, 2k sectors per track and 2048 bytes per sector. Disk rotates with 3000 rpm, and seek time is 10 ms then find the disk access time?	4	CO4	L3
Q.3. (a)	Suppose six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order) is given, how would first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)?	4	CO3	L3
(b)	Discuss the paging technique and explain how it addresses fragmentation problems. Consider logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. **Solve the following queries.**	4	CO1	L2

	I. How many bits are there in the logical address? II. How many bits are there in the physical address?			
Q.4. (a)	Could you simulate a multilevel directory structure with a single-level directory structure in which arbitrarily long names can be used? If your answer is yes, explain how you can do so, and contrast this scheme with the multilevel directory scheme. If your answer is no, explain what prevents your simulation's success. How would your answer change if file names were limited to seven characters?	4	CO2	L3
(b)	Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 How many page faults would occur for the following mentioned replacement algorithms, assuming three frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each. Illustrate all steps briefly. I. LRU replacement II. Optimal replacement	4	CO3	L4
Q.5. (a)	Consider the set of 5 processes whose arrival time and burst time are given below: <table><tr><td>Process ID</td><td>Arrival Time</td><td>Burst Time</td><td>Priority</td></tr><tr><td>P1</td><td>0</td><td>4</td><td>2</td></tr><tr><td>P2</td><td>1</td><td>3</td><td>3</td></tr><tr><td>P3</td><td>2</td><td>1</td><td>4</td></tr><tr><td>P4</td><td>3</td><td>5</td><td>5</td></tr><tr><td>P5</td><td>4</td><td>2</td><td>5</td></tr></table> Calculate the average waiting time and average turnaround time, using preemptive priority scheduling and round robin algorithm (higher number represents higher priority and time quantum is 3-time unit).	4	CO1	L4
(b)	Consider a disk queue with requests for I/O to blocks on cylinders are: 82, 170, 43, 140, 24, 16, 190. The head is initially at cylinder number 50. Apply SSTF, C-SCAN, LOOK and C-LOOK scheduling algorithms. The cylinders are numbered from 0 to 199. Find out total head movement (in number of cylinders) incurred while servicing these requests.	4	CO4	L4
Q.6. (a)	Write short note on following using suitable example: I. CIA (Confidentiality, Integrity, Availability) II. Denial of Service	4	CO5	L3
(b)	In semaphores, which three kinds of problem can be solved with explanation? Consider a non-negative counting semaphore S. The operation P(S) decrements S, and V(S) increments S. During an execution, 20 P(S) operations and 12 V(S) operations are issued in some order. Find out the largest initial value of S for which at least one P(S) operation will remain blocked. Justify it with proper explanation.	4	CO3	L4

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